

ATPL EASA STUDY GUIDE | 021 - AIRFRAME & SYSTEMS

Chapter 6: Pressurisation Systems (Regimes & Outflow Valves)

Syllabus Source: CAE Oxford (pp. 217-234) | Pass Target: >= 90% AviationExam

1. PRESSURIZATION PRINCIPLES & DIFFERENTIAL PRESSURE LIMITS

Passenger transport aircraft fly in the thin upper atmosphere (FL 300 to FL 430) for engine fuel efficiency and to avoid adverse weather. The cabin must be pressurized to maintain physiological oxygen saturation without requiring oxygen masks:

- Cabin Altitude: The equivalent altitude inside the passenger cabin corresponding to the internal pressure. Under CS-25/FAR-25, normal maximum cabin altitude is 8,000 ft (ambient pressure ~750 hPa). Modern airliners (A350, B787) maintain 6,000 ft cabin altitude due to composite fuselage strength.
- Differential Pressure (Delta-P): The difference between internal cabin pressure (P_{int}) and external atmospheric ambient pressure (P_{amb}): $\Delta P = P_{int} - P_{amb}$. Maximum design differential pressure is typically 8.0 to 9.0 psi (aluminum airframes) and up to 9.4 psi (composite airframes).
- Cabin Rate of Climb / Descent: Rate of change of cabin pressure expressed in feet per minute. Normal passenger comfort limits: +500 ft/min during climb; -300 ft/min during descent.

2. OUTFLOW VALVE & PRESSURE REGULATION ARCHITECTURE

A common misconception is that pressurization is controlled by varying incoming air supply. In reality, incoming air conditioning pack mass flow is CONSTANT; cabin pressure is regulated entirely by controlling the EXHAUST RATE of air overboard:

Valve / Unit	Engineering Location	Operational Function & Pressure Limit
Outflow Valve (OFV)	Lower rear fuselage (under pressure bulkhead).	Modulates exhaust area from fully open (ground) to nearly closed (cruise). Controlled by dual digital pressurization controllers.
Positive Relief Valves	Fuselage pressure shell.	Spring-loaded mechanical safety valves. Open automatically if cabin differential pressure exceeds maximum structural limit (e.g. 8.6 to 9.1 psi) to PREVENT FUSELAGE EXPLOSION!
Negative Relief Valves (Inward)	Fuselage pressure shell.	Inward-opening spring valves. Open if ambient pressure exceeds cabin pressure (e.g. in rapid emergency dive), preventing fuselage structural crushing (max negative diff ~ -0.5 psi).
Dump Valve (Safety Valve)	Electrically or pneumatically signaled.	Dumps all cabin pressure to ambient when aircraft touches down on runway or when pilot selects DUMP switch.

[!] EXAM TRAP (AVIATIONEXAM) - Cabin Altitude Warning & Passenger Oxygen Trigger Points

- 8,000 ft: Normal maximum cabin altitude in commercial cruise. - 10,000 ft: CABIN ALTITUDE WARNING! Audio horn / master warning light triggers. Flight crew must immediately don oxygen masks (100% oxygen) and initiate emergency descent if uncontrollable! - 14,000 ft: AUTOMATIC PASSENGER OXYGEN MASK DEPLOYMENT. Chemical generator masks drop automatically from overhead passenger service units (PSU).

3. DECOMPRESSION CATEGORIES & PHYSIOLOGICAL RISKS

Loss of cabin pressurization can occur through seal failure, door release, or structural breach:

- **Slow / Gradual Decompression:** Loss of pressure takes longer than 10 seconds. Extremely insidious: flight crew may succumb to hypoxia without recognizing symptoms unless instrument warnings are monitored.
- **Rapid Decompression:** Loss of pressure occurs between 1 and 10 seconds. Accompanied by loud bang, rush of air toward breach, sudden fogging (moisture condensation as temperature drops), and drop in cabin temperature.
- **Explosive Decompression:** Loss of pressure occurs in LESS THAN 0.5 SECONDS (faster than lungs can decompress without damage). Lungs suffer trauma if pilot holds breath; Time of Useful Consciousness (TUC) is slashed by 33% to 50%!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Cabin Altitude & Outflow Valve R**

Golden Rules of Pressurization: 1. Max normal cabin altitude: 8,000 ft (CS-25 transport standard). 2. Cabin Altitude Warning Horn: Sounds at 10,000 ft cabin altitude. 3. Passenger Oxygen Masks Drop: Automatically deploy at 14,000 ft cabin altitude. 4. Outflow Valve Position: Modulates from full open on ground to almost closed at FL410 to maintain target differential pressure (typical 8.5 psi).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: How is cabin pressure regulated in a modern jet transport aircraft during normal flight?**

- [A]: By varying the engine bleed air mass flow rate entering through the packs
- [B]: By modulating the position of the outflow valve(s) to control the rate of air escaping overboard
- [C]: By opening and closing cabin passenger doors
- [D]: By throttling the engine fuel pumps

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The air conditioning packs deliver a constant mass flow of fresh air. The digital pressurization controller commands the outflow valve to open or close, regulating the rate of air exhausted overboard.

Q2: At what cabin pressure altitude does the master cabin altitude warning horn activate under commercial tra

- [A]: 8,000 ft
- [B]: 10,000 ft
- [C]: 14,000 ft

- [D]: 18,000 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. At 10,000 ft cabin altitude, an intermittent audio horn and visual warning alert the pilots to don oxygen masks immediately. (Passenger masks deploy automatically at 14,000 ft).

Q3: What is the primary operational purpose of inward-opening negative relief valves on a pressurized aircraft

- [A]: To prevent cabin overpressure in cruise
- [B]: To prevent external atmospheric ambient pressure from exceeding internal cabin pressure, which would crush the fuselage inward
- [C]: To allow passenger cabin smoke evacuation
- [D]: To supply ram air to the engines

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Aircraft fuselage structures are designed as pressure vessels to withstand internal pressure (tension). They have poor resistance to external pressure (compression/crushing). Negative relief valves open inwards to equalize pressure if ambient exceeds cabin pressure.

Q4: Why does explosive decompression reduce the Time of Useful Consciousness (TUC) by 33% to 50% comp

- [A]: The blood freezes instantly
- [B]: Ambient pressure drops faster than alveolar partial pressure, causing reverse diffusion of oxygen from the blood into the lungs
- [C]: Adrenaline blocks hemoglobin
- [D]: Heart rate drops to zero

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. In explosive decompression, alveolar oxygen pressure drops instantly below blood capillary oxygen pressure. Oxygen diffuses backwards out of the venous blood into the lungs, draining oxygen from the brain rapidly.